

A disproof of Conjecture 00000007751

A quadratic rational map over $\mathbb{Q}$ with twelve rational preperiodic points, so $\text{MSD}(1, 2) \neq 9$

1 The conjecture

Conjecture 00000007751 reads (English version; the Chinese version states the same conjecture):

Definition: For a rational map f of degree $d \geq 2$ over a field K , a preperiodic point has finite orbit; the Morton–Silverman uniform boundedness conjecture asserts a uniform bound on preperiodic points of degree d maps of $\mathbb{P}^N(K)$, depending only on N and d , with $\text{MSD}(N, d)$ the minimal such bound. *Conjecture:* For $N = 1$, $d = 2$, $K = \mathbb{Q}$, $\text{MSD}(1, 2) = 9$, attained by a unique map; more generally $\text{MSD}(N, d)$ is dominated by the exponential expression $(d^{2N+2} - 1)/(d - 1)$.

Reading. Here $\text{MSD}(1, 2)$ (over $K = \mathbb{Q}$) is the least uniform bound, taken in $\mathbb{N}_\infty = \{0, 1, 2, \dots\} \cup \{\infty\}$, for the number of $\mathbb{Q}$ -rational preperiodic points in $\mathbb{P}^1(\mathbb{Q})$ of a degree-2 rational map defined over $\mathbb{Q}$; it equals ∞ if no finite uniform bound exists. The statement is explicitly about *rational maps*, not only polynomials. The conjecture is the conjunction of three clauses:

- (i) $\text{MSD}(1, 2) = 9$;
- (ii) this value is attained by a unique map;
- (iii) in general $\text{MSD}(N, d)$ is dominated by $(d^{2N+2} - 1)/(d - 1)$.

We refute clause (i), which refutes the conjunction. We do **not** address the uniqueness clause (ii) or the general domination clause (iii). For $N = 1$, $d = 2$ the expression in (iii) equals $(2^4 - 1)/(2 - 1) = 15$, which is consistent with the twelve points exhibited below; nothing here contradicts (iii).

2 Conventions

These are the standard definitions of a degree- d rational map of $\mathbb{P}^1$ and of preperiodic points, and they are exactly the ones used in the Lean file.

- $\mathbb{P}^1(\mathbb{Q})$ is modelled as $\mathbb{Q} \cup \{\infty\}$ (in Lean, `Option Q`): the point $a \in \mathbb{Q}$ is $[a : 1]$ and $\infty = [1 : 0]$ (Lean `none`).
- A *rational map of degree d* over $\mathbb{Q}$ is $\varphi = p/q$ with $p, q \in \mathbb{Q}[X]$ coprime and $\max(\deg p, \deg q) = d$.
- φ acts on $\mathbb{P}^1(\mathbb{Q})$ by the homogeneous formula $[x : y] \mapsto [F(x, y) : G(x, y)]$ with $F = Y^d p(X/Y)$, $G = Y^d q(X/Y)$, i.e.

$$[a : 1] \mapsto [p(a) : q(a)], \quad [1 : 0] \mapsto [\text{coeff}_d p : \text{coeff}_d q],$$

where $[u : v] = u/v$ if $v \neq 0$ and $[u : 0] = \infty$.

- A point x is *preperiodic* if its forward orbit $\{\varphi^n(x) : n \geq 0\}$ is finite, exactly as in the statement (“a preperiodic point has finite orbit”).

Lemma 1 (well-definedness; Lean `RatMap.act_wellDefined`). *If $d \geq 1$ and $\varphi = p/q$ has degree d , then $(p(a), q(a)) \neq (0, 0)$ for all $a \in \mathbb{Q}$, and $(\text{coeff}_d p, \text{coeff}_d q) \neq (0, 0)$. Hence the formula above never produces $[0 : 0]$.*

Proof. Coprimality gives $u, v \in \mathbb{Q}[X]$ with $up + vq = 1$; evaluating at a rules out $p(a) = q(a) = 0$. If, say, $\deg q = d \geq \deg p$, then $q \neq 0$ (otherwise $d = \deg q = 0$), so $\text{coeff}_d q$ is the leading coefficient of q and is nonzero; symmetrically if $\deg p = d$. $\square$

3 The witness

Let

$$f(z) = \frac{P(z)}{Q(z)} = \frac{-2z^2 - 3z - 1}{2z^2 - z} = -\frac{(2z+1)(z+1)}{z(2z-1)}.$$

Both P and Q have degree 2, and they are coprime by the Bézout identity

$$(4X - 3)P + (4X + 5)Q = 3$$

(expand: $(4X - 3)P = -8X^3 - 6X^2 + 5X + 3$ and $(4X + 5)Q = 8X^3 + 6X^2 - 5X$). Alternatively, the resultant of P and Q is $12 \neq 0$. So f is a rational map of degree 2 over $\mathbb{Q}$; it is not a polynomial. At infinity, $f(\infty) = [\text{coeff}_2 P : \text{coeff}_2 Q] = [-2 : 2] = -1$.

Let

$$S = \{\infty, -1, 0, -\frac{5}{2}, -\frac{2}{5}, -\frac{1}{6}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, -\frac{3}{2}, -\frac{1}{3}, 2\} \subset \mathbb{P}^1(\mathbb{Q}),$$

twelve distinct points. Direct evaluation gives:

x	$P(x)$ (resp. $\text{coeff}_2 P$)	$Q(x)$ (resp. $\text{coeff}_2 Q$)	$f(x)$
∞	-2	2	-1
-1	0	3	0
0	-1	0	∞
$-5/2$	-6	15	$-2/5$
$-2/5$	$-3/25$	$18/25$	$-1/6$
$-1/6$	$-5/9$	$2/9$	$-5/2$
$-1/2$	0	1	0
$1/2$	-3	0	∞
$-1/4$	$-3/8$	$3/8$	-1
$-3/2$	-1	6	$-1/6$
$-1/3$	$-2/9$	$5/9$	$-2/5$
2	-15	6	$-5/2$

Thus f has the two 3-cycles

$$\infty \rightarrow -1 \rightarrow 0 \rightarrow \infty, \quad -\frac{5}{2} \rightarrow -\frac{2}{5} \rightarrow -\frac{1}{6} \rightarrow -\frac{5}{2},$$

and the tails $-\frac{1}{2} \rightarrow 0, \frac{1}{2} \rightarrow \infty, -\frac{1}{4} \rightarrow -1, -\frac{3}{2} \rightarrow -\frac{1}{6}, -\frac{1}{3} \rightarrow -\frac{2}{5}, 2 \rightarrow -\frac{5}{2}$. In particular $f(S) \subseteq S$.

Lemma 2 (Lean `isPreperiodic_of_mem`). *Let $g : X \rightarrow X$ and let $S \subseteq X$ be finite with $g(S) \subseteq S$. Then every $x \in S$ is preperiodic for g .*

Proof. By induction on n , $g^n(x) \in S$ for all $n \geq 0$. So the forward orbit of x is contained in the finite set S , hence finite. $\square$

4 The refutation

Theorem 3. $\text{MSD}(1, 2) \geq 12$. *In particular $\text{MSD}(1, 2) \neq 9$, and the conjecture is false.*

Proof. By Lemma 2 and $f(S) \subseteq S$, all twelve points of S are $\mathbb{Q}$ -rational preperiodic points of the degree-2 map f . Hence every uniform bound $B \in \mathbb{N}_\infty$ (a B with $\#T \leq B$ for every degree-2 map g over $\mathbb{Q}$ and every finite set T of $\mathbb{Q}$ -rational preperiodic points of g) satisfies $12 = \#S \leq B$. So 12 is a lower bound of the set of uniform bounds and $12 \leq \text{MSD}(1, 2)$, the infimum of that set. (If no finite bound exists, the only bound is ∞ and $\text{MSD}(1, 2) = \infty \neq 9$.) Since $12 > 9$, $\text{MSD}(1, 2) \neq 9$. Equivalently, it is false that every degree-2 rational map over $\mathbb{Q}$ has at most nine $\mathbb{Q}$ -rational preperiodic points in $\mathbb{P}^1(\mathbb{Q})$. $\square$

Remark 4. The refutation does not depend on whether ∞ is counted: the eleven finite points $S \setminus \{\infty\}$ are already preperiodic, and $11 > 9$ (Lean `exists_degree_two_eleven_affine_preperiodic`). We claim only that the points of S are preperiodic; we do not claim that S is the full set of rational preperiodic points of f , nor anything about the exact value of $\text{MSD}(1, 2)$ beyond $\text{MSD}(1, 2) \geq 12$.

Context. The setting matches the one studied in the literature. The abstract of Benedetto, Chen, Hyde, Kovacheva and White, arXiv:1312.0491 (<https://arxiv.org/abs/1312.0491v2>) begins: “Let $f \in Q(z)$ be a polynomial or rational function of degree 2. A special case of Morton and Silverman’s Dynamical Uniform Boundedness Conjecture states that the number of rational preperiodic points of f is bounded above by an absolute constant.” We use this only as context; the proof above is self-contained and relies on no result from that paper.

5 Lean 4 formalization

The directory `lean/` contains a Lean 4 project (Lean `v4.33.1`, Mathlib `v4.33.1`). The single source file `Conjecture7751/Basic.lean` (namespace `C7751`) defines `RatMap d` (fields `p q : ℚ[X]`, `IsCoprime p q`, `max p.natDegree q.natDegree = d`), the action `RatMap.act` on `Option ℚ` by the homogeneous formula, `IsPreperiodic g x := (Set.range fun n => g^[n] x).Finite`, and

$$\text{MSD } d := \inf\{B \in \mathbb{N}_\infty \mid \forall f : \text{RatMap } d, \forall S, (\forall x \in S, \text{IsPreperiodic } f.\text{act } x) \rightarrow \#S \leq B\},$$

with S ranging over `Finset (Option ℚ)`. Main results:

Lean name	Statement
<code>RatMap.act_wellDefined</code>	for $d \geq 1$: $(p(a), q(a)) \neq (0, 0)$ for all a , and $(\text{coeff}_d p, \text{coeff}_d q) \neq (0, 0)$
<code>isPreperiodic_of_mem</code>	a point of a finite forward-invariant set is preperiodic
<code>P_Q_coprime</code>	$\text{IsCoprime } P \ Q$ (via the Bézout identity above)
<code>P_natDegree, Q_natDegree</code>	$\deg P = \deg Q = 2$
<code>f_act_some, f_act_none</code>	$f([a : 1]) = [-2a^2 - 3a - 1 : 2a^2 - a]$; $f(\infty) = -1$
<code>S_card</code>	$\#S = 12$
<code>S_invariant</code>	$\forall x \in S, f(x) \in S$
<code>exists_degree_two_twelve_preperiodic</code>	some $g : \text{RatMap } 2$ and T with $\#T = 12$ and every $x \in T$ preperiodic for g
<code>exists_degree_two_eleven_affine_preperiodic</code>	some $g : \text{RatMap } 2$ and $T : \text{Finset } \mathbb{Q}$ with $\#T = 11$ and every some $a, a \in T$, preperiodic
<code>twelve_le_MSD</code>	$(12 : \mathbb{N}_\infty) \leq \text{MSD } 2$
<code>MSD_ne_nine</code>	$\text{MSD } 2 \neq 9$
<code>not_all_le_nine</code>	$\neg \forall g : \text{RatMap } 2, \forall T, (\forall x \in T, \text{IsPreperiodic } g.\text{act } x) \rightarrow \#T \leq 9$

There is no sorry, no native_decide and no added axiom. #print axioms for MSD_ne_nine, not_all_le_nine, exists_degree_two_eleven_affine_preperiodic and RatMap.act_wellDefined reports only propext, Classical.choice and Quot.sound. To check:

```
cd lean
lake exe cache get
lake build
```